

Enabling Ubiquitous High-Fidelity Cardiac Auscultation via Integration of Smartphone and 3D Printing Technologies

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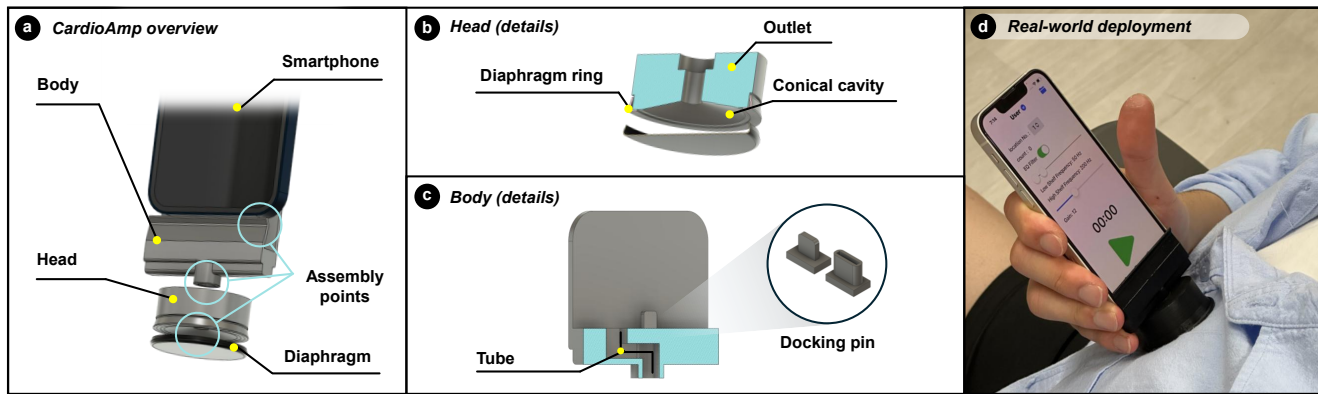


Figure 1: Overview of CardioAmp. (a) CardioAmp incorporates smartphone and 3D printing technologies for cardiac auscultation. (b) The head is structurally designed to capture clear low-frequency cardiac sounds. (c) The body acts as a bridge between the head and the smartphone. (d) CardioAmp enables users to perform high-quality cardiac auscultation anytime, anywhere.

Abstract

We present CardioAmp, a novel cardiac auscultation system that integrates a 3D-printed accessory with a smartphone to enable ubiquitous, high-fidelity cardiac sound collection. Specifically, to supplement the smartphone's lack of sensitivity to low-frequency cardiac sounds, we design each component of the accessory, called head and body, to incorporate structures that effectively capture and transmit heart sounds and murmurs to the smartphone's microphone. Our experimental results show that CardioAmp can improve the quality of cardiac sound recordings compared to standalone smartphone auscultation, thus opening new opportunities for daily self-monitoring and early cardiovascular disease detection.

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CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Applied computing** → **Consumer health**.

Keywords

Cardiac auscultation, Digital stethoscope, 3D printing

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1 INTRODUCTION

Cardiac auscultation is one of the simplest and most effective methods for early detection and monitoring of cardiovascular disease, a leading cause of death worldwide. However, despite its importance, auscultation is typically conducted only in clinical environments

(e.g., hospitals). This sporadic examination often misses opportunities to identify early signs of deterioration during daily life.

As an initial step toward enabling cardiac auscultation beyond clinical facilities, several efforts have focused on developing digital stethoscopes capable of capturing high-fidelity cardiac sounds. However, none of these solutions are yet suitable for daily auscultation due to their dependence on specialized devices or their limited recording quality. For example, many commercial digital stethoscopes [2, 4, 5] have been introduced with unique hardware designs and diverse software functionalities (e.g., noise cancellation) aimed at enhancing the quality of cardiac sound recordings. However, their high cost (typically above USD 300) hinders their widespread adoption. 3D-printed stethoscopes [6, 8, 10, 15–17] offer a low-cost alternative, which demonstrates auscultation performance even comparable to commercial devices. Nevertheless, their reliance on additional hardware components, such as external microphones and circuits, serves as a barrier to practical use. Other studies [7, 11–14] have leveraged smartphones equipped with built-in microphones as standalone auscultation systems. While these approaches allow cardiac sound collection in everyday contexts, they compromise auscultation quality since smartphone microphones are optimized for speech rather than low-frequency cardiac sounds.

In this study, we overcome these limitations by combining the strengths of 3D-printed and smartphone-based stethoscopes. As shown in Figure 1, we introduce CardioAmp, which enables cardiac sound collection (i) ubiquitously, using a smartphone’s built-in microphone, and (ii) with high quality, through a 3D-printed auscultation accessory.

2 CardioAmp Design

One major requirement to enable ubiquitous high-fidelity cardiac auscultation is to capture cardiac sounds, such as heart sounds and murmurs, which contain frequency components in the range of 20–1,000 Hz, with high fidelity. However, commercial smartphone microphones lack the required sensitivity, as they are primarily tuned for the frequency range of human speech.

CardioAmp addresses this challenge with a 3D-printed auscultation accessory that acts as a physical amplifier for cardiac sounds. It consists of two components: a *head* and a *body* (see Figure 1). The accessory acts as a physical amplifier for low-frequency. The head closely resembles commercial stethoscope chest pieces to effectively capture the mechanical vibrations generated by heartbeats. For example, its bottom rim, called a diaphragm ring, is designed to hold a thin film diaphragm with a diameter of 44 mm, which is typical for adult cardiology stethoscopes [3]. When placed against the skin, the diaphragm converts subtle mechanical vibrations from the contacted area into acoustic signals. These signals are then directed toward an outlet on the opposite side. In particular, the head’s conical cavity helps funnel the sound waves from the wide diaphragm to the narrow opening, reducing reflection and energy loss while preserving low-frequency components. It should be noted that the basic structure of the head is designed based on an open-source 3D-printed stethoscope system [9].

As illustrated in Figure 1(c), the body contains a tube that guides sound waves from the head’s outlet to a smartphone’s microphone. It attaches securely to the bottom of the smartphone via its clip-like

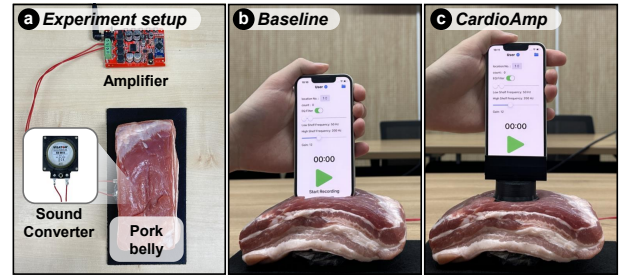


Figure 2: Experimental setup.

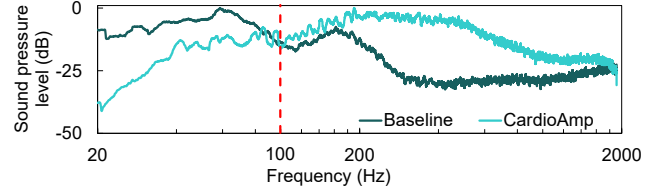


Figure 3: Frequency response of CardioAmp, measured with band-limited Gaussian white noise. Sound pressure levels were normalized to the maximum.

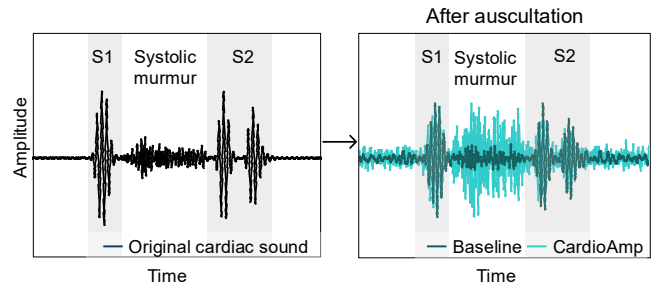


Figure 4: Cardiac auscultation performance of CardioAmp. We used a cardiac sound recording, which includes eject systolic murmurs with split S2.

structure and docking pin. In particular, the docking pin fits into the smartphone’s charging port, ensuring precise alignment between the tube and the microphone. The opposite end of the tube is tightly assembled with the head’s outlet, thereby forming a bridge between the head and the smartphone. Notably, we designed the tube to be short (3.3 cm in length) and wide (6–10 mm in diameter) to transfer low-frequency waves with minimal loss.

3 Evaluation

We conducted experiments to evaluate how effectively CardioAmp captures low-frequency cardiac sounds. As shown in Figure 2(a), we emulated real-world cardiac sound propagation using a sound converter placed beneath a pork belly with a thickness of 40 mm. The converter played band-limited (20–2000 Hz) Gaussian white noise and a cardiac sound recording from a public dataset [1]. We then recorded the transmitted signals using a *baseline* system (smartphone only) and CardioAmp (see Figure 2(b) and (c)).

Figure 3 demonstrates that CardioAmp achieves substantially higher amplitude than the baseline system across the 100–2,000 Hz

frequency range. This improvement is due to the 3D-printed auscultation accessory specifically designed to amplify low-frequency sounds. In particular, the enhanced sensitivity increases the likelihood of capturing clear heart murmurs. As shown in Figure 4, the baseline system can record heart sounds, called S1 and S2, clearly but fails to capture murmurs with sufficient clarity. In contrast, CardioAmp reliably records both heart sounds and murmurs through the low-frequency amplification. These results confirm that CardioAmp can provide high-quality cardiac sound recordings suitable for detecting cardiac abnormalities. Note that CardioAmp still exhibits slight intensity degradation in the 20–100 Hz range, which includes important components of heart sounds (see Figure 3). As part of our future work, we plan to explore refinements to the accessory structure to improve amplification across both heart sounds and murmurs.

4 CONCLUSION

In this paper, we presented CardioAmp, a novel cardiac auscultation system that combines 3D-printed auscultation components and smartphones to capture high-quality cardiac sounds. While CardioAmp significantly improves recording quality over standalone smartphone auscultation, challenges remain in noise robustness and usability. In future work, we will explore hardware and software solutions to address such challenges.

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